FLOODGAUGE · PROJECT PLAN

FloodGauge project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/11-plan.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Initial draft for the kit 1.9.1 upgrade

FloodGauge project plan

FloodGauge is at stage P3 (constructable design, TRL 3, with detailed CAD for three concept options). The next gate is Amish's choice between option A and option B (FLG-OPT-001, FLG-DDR-004); P4 to P6 stay on hold while the portfolio stage cap is P3. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Design lead	PRB, URS	Done at v0.6 and v0.1; co-design sessions not held
STEM check	Design lead	STM, stem_check.py	Done; independent review recorded
Concept options	Design lead	OPT, DDR-004, options picture	Done; choice awaits Amish; options picture rendered by the coordinator
Requirements	Design lead	PRS, RTM	Done; PR-001 to PR-019
Calculations	Design lead	CAL, sizing.py	Done at v0.7
CAD	Design lead	model.py for A, B and C, STEP, STL, drawings	Done at D3
Build plan	Design lead	BLD, pictures	Done for option A (paper only)
Regulation, feasibility, risk, market	Design lead	REG, FEA, RSK, MKT	Done at v0.1; items "to assess" open
Pilot partner	Project lead	A host city or county agency	Not started
Prototype build and test	Builder	Hardware, TST	On hold (TRL 4, not started)

Resources needed

Table 2. Resources. Estimates are for planning only; no purchase is planned at TRL 3.

Resource	Deliverable	Effort	Estimated cost (USD)
Pilot partner liaison	Site, permits (REG G9, G10)	2 to 4 weeks part-time	To assess
Radar module data sheet and quote	Close STM S3, R-1	1 day	0
Parts for one prototype	Option chosen by Amish	1 build, 2 to 3 days	205.50 for option A (BOM, indicative)
Metal brake and workshop time	Hat covers, brackets	1 day	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0	STEM check, concept options, REG, FEA, PLN	Reached 2026-10-31
P1	URS, PRS, RSK, MKT	Reached 2026-10-31
P2	RTM, CAL	Reached 2026-10-31
P3	BLD, DEC, constructable design, detailed CAD for every option, design FMEA	Reached 2026-10-31
P4	Prototype built and tested (TST)	On hold (stage cap)
P5	Relevant-environment test	On hold
P6	Released precis v1.0, drawings at a lettered revision	On hold

Next steps and decision requested

Decision requested from Amish: pick option A or option B, or ask for the hybrid of FLG-DDR-004 option 3 (radar over the street plus a hung pressure transducer in the basin). The recommended next step after that is to read the radar module data sheet, approach the pilot partner, and then plan the TRL 4 build for the chosen option when the portfolio cap allows.

References

Sources for every cited fact are in FLG-REG-001 and FLG-MKT-001 and the problem statement FLG-PRB-001.